

THE FUTURE OF SCHOLARSHIP

A DISSERTATION
SUBMITTED TO THE DEPARTMENT OF PSYCHOLOGY
AND THE COMMITTEE ON GRADUATE STUDIES
OF STANFORD UNIVERSITY
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF
DOCTOR OF PHILOSOPHY

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April 2025

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I certify that I have read this dissertation and that, in my opinion, it is fully adequate in scope and quality as a dissertation for the degree of Doctor of Philosophy.

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(Carl Friedrich Gauss)

Approved for the Stanford University Committee on Graduate Studies

Abstract

I'm an abstract right here, look at me!

Dedication

Acknowledgments

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Chapter 1

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Chapter 2

Introduction: Towards formal theories and computational models of evolving referents to unfamiliar targets

2.1 Pre intro on the wonders of language

Language means different things to different people within different areas of linguistics. There's language as a cultural artifact, language as a symbolic system, language as a communication modality and technology, language as a process.

Here, I focus on language as a process for communicating that is used in a variety of contexts. All the language as a reified thing is really made up of many many individual cases of language use in specific contexts. There is no semantics, only conventional pragmatics.

One of the really cool things about language is its extensibility. The combinatorial feature of this is well noted – we can produce and comprehend sentences that we have never produced or heard before, even ones that perhaps no one has produced or heard before. Beyond the compositionality portion, we can also stretch the meanings of words and constructions to match communicative needs. That's one of the ways language evolves, as new words and constructions come to be to accomodate new

scientific and cultural developments that need referring to. However, this also plays out at a finer grained scale – many neologisms or creative usages do not become fixed in language as a whole but are fleeting – useful in the moment to serve a communicative need, or perhaps sticking around as a joke between small groups.

How are we able to use language in these flexible adaptive ways? How do we develop understandings when the meaning isn't codified? One key aspect is that language is used in a context, so there the information of the utterance is not just in the words, but also in the extra-textual information – the priors as it were.

Similar to language as a whole, pragmatics means different things to different people. Here, we take the pragmatics wide view (CITATION) as any context - language interaction that enhances meaning beyond the most typical meaning.

“language in context” what the pressures on utterances – informativity, availability, how language & context interact – non-“literal” uses of language, information value in the extra-textual or sub-textual ways pragmatics wide – conveying information beyond the literal meaning < – what do you call this because if I say pragmatics some people think scalar implicature (or implicature generally)

flexibility of language use (both production and comprehension) – add new meanings, new words, adapt to circumstances

interested in production and processing (processing is easier) with these

language is a very rich code and is an avenue for seeing various human skills because of the shared suite of tools

Language as a tool for communication & the role of context & adaptation to context in language use: * broad topic * “nothing is all that special” <– no theoretic primacy to any particular form including conversation or even language (compared with other modalities) * messy – easy to observe, harder to characterize, and harder still to computationally model * situated between a number of different fields * language evolution / cultural evolution / how new usages enter a lexicon <– difficult to observe IRL * pragmatics (but this means different things to different people) * “mechanics” of processing and production * “inference” where are we on the language <-> wordplay spectrum * conversation & social interaction

%TODO FOR LATER an example can strengthen this kind of abstract info and make it more

%TODO FOR LATER can you say a bit about why we'd want to do this? other than "because science

%TODO FOR LATER be more explicit about the Philosophy of science backing here

Language is an amazing human technology that supports much of human culture by being an

Much linguistic creativity is ephemeral, occurring at at a single time within a single c

Whether descriptions are fleeting or lasting, the core observation is the same: humans a

How can we explain and predict these phenomena? Following \citet{meehl1990}, we will pre

2.2 Here we

How do we use language in these adaptive contexts?

As a practical matter, it's difficult to study this in the world at scale, so experimental models are commonly used. One issue with developing formal models of human linguistic behavior is that the data are rich and the possibilities are vast. How can one take rich, messy interaction data and tame it into meaningful features that are tractable to model and predict?

While language is used for many communicative acts, a core component is reference – reaching agreement on the target that goes with a portion of linguistic material. What do those words index? Sometimes this can be isolated, as in reference games, but it is also a pre-requisite for various other communicative acts. It's hard to negotiate or coordinate when one doesn't agree on the targets of the NPs.

Reference can be studied in a class of experiments called reference games, which involve the production or interpretation of a utterance to refer to a specific object among a (usually fixed) set of objects. Reference games offer control in that the

experimenter can control the context of objects, which the desired target is, and depending on the set-up, potentially the phrase or the response. Depending on the set-up, one can get a number of dependent measures. In elicitation studies, one gets the referring expressions (and potentially the time course and prosody); in interpretation studies, one gets the selecting behavior of what target is selected and when. Additional behavioral measures such as looking, incremental reaching, or clarification are also sometimes collected.

As it will be the main focus on this dissertation, we also introduce a special kind of reference game: the iterated reference game. Here, a pair or group of participants do a series of referring tasks with similar contexts and targets on each trial. Typically, the targets and context are designed such that initial reference is non-trivial, with targets being low codeability (hard to name) or at least hard to distinguish from their competitors. Over repetition, the group builds up some context that can be used to inform future references. Iterated reference games serve as a useful way of eliciting context-dependent creative language use & convention formation and thus is a useful testbed for studying different dimensions of pragmatics.

In the rest of this introductory chapter, we review some of the relevant background literature, including some of the relevant key questions and theories. We will start by surveying the literature related to pair communication and reference which consists primarily of verbal theories that document some observed phenomena, but do not cache out as testable predictions. We will then zoom out to three more established larger frameworks of language more broadly – efficiency, information theory, and Bayesian reasoning. These are not mutually exclusive, or really independent. Armed with the prior literature, and the large frameworks, we will look at what the weak points are where more empirical evidence could usefully inform the debate and eventually allow for the construction of satisfying, testable computational models.

Some questions: * do we do framework first or phenomena first? maybe actually framework first? * which frameworks? * what doesn't fit?

2.3 Frameworks

2.3.1 1: efficiency (& info theory)

One unifying framework gaining traction in psycholinguistics is efficiency, the idea that language and language use is under pressure to support efficient communication by maximizing the ratio of relevant information transmitted to effort. Efficiency is a high level framework that requires a number of linking assumptions to render it testable against data; however, comparisons of attested language use to counterfactual options can bound what assumptions are needed for parsimony.

Efficiency is thought to arise from trade-offs between communicative expressivity and some combination of learnability and ease of production (Kirby et al., 2015; Piantadosi et al., 2012).

Evidence for efficiency comes from the argument that features of language are distributed much more closely to the Pareto frontier than would be expected by chance. A historically well-known example is that word frequencies follow a power-law distribution, which Zipf (1949) explains in terms of a “principle of least effort”, although note that power-law distributions are common across domains and generated by a variety of processes (Piantadosi, 2014). Stronger evidence comes from the lexical partitioning of subdomains such as color, number, and kinship terms, where the distribution of systems falls on the frontier between complexity (number of terms) and informativity (how many bits each term provides) (Gibson et al., 2019; Kemp et al., 2018; Zaslavsky et al., 2018). Syntactic features of language such as harmonic word order and dependency length also appear to be optimized for increased expressivity with minimized processing effort (Gibson et al., 2019; J. Hawkins, 1995).

Efficiency arguments are based on the language artifacts of grammars and transcripts, but efficiency pressures act on language use as a process, not language as a static code (Gibson et al., 2019). Thus efficiency can be seen as imposing a joint constraint on the entire communicative process: minimizing the total time and effort involved in going from an idea in one person’s head to a sufficiently close idea in another person’s head. A corollary of this framing is that shorter utterances (as measured in syllables or clock-time) are not always efficient if they take longer to produce or parse.

One issue with efficiency is that as a framework, it lives and dies by the auxiliary theories (NEVER BE WRONG CITATION) and so for various things the question will be what sort of auxiliary theories would be needed and how do we feel about those? As an illustration, we look at the issues of ambiguity and redundancy in reference expressions.

Redundancy and over-informative referring expressions

How do we reconcile the evidence that language is efficient with violations of efficiency in language use (“look at the yellow banana”) and the ubiquity of ambiguous utterances?

Terms such as “redundant” and “over-informative” are commonly used to describe situations where people produce modifiers that do not restrict the extension of a noun phrase (e.g. “blue cup” when only one cup is salient) (Rubio-Fernandez et al., 2021). People do produce these non-restricting modifiers in reference tasks, especially for color, but this behavior seems to run counter to the idea of efficient language use. Is this a contradiction?

Formalizing claims of redundancy or over-informativity requires a definition of what would be minimally informative, which in turn depends on a commitment to a fully specified semantic-pragmatic system. For instance, if specificity implicatures are within the option space, are those calculated before or after informativeness is measured (**bergen?**)? One could sidestep the thorny theoretical by empirically measuring the information content of different utterances by how they shift the entropy of the distribution of inferred meanings (**degen20200406?**), but this does not scale up well.

The flip side of “redundancy” is ambiguity: many, many utterances are ambiguous. In general, strong contextual factors render the ambiguity a non-issue (Piantadosi et al., 2012), but this means we can’t judge language outside of the physical and social context it is used in. Determining what is efficient language use in context requires not just analyzing phrases and their alternatives, but also how long utterances take to generate and comprehend, which may be highly contingent on contextual factors and conversational history.

The idea that utterances should have “just enough” information has inspired a

wealth of empirical research into what utterances people produce and what utterances people comprehend. By comparing these two halves of language use, we can determine how calibrated utterances are to what other people will understand.

2.3.2 Framework 2: Bayesian

should mention briefly that there's larger resource rational and Bayesian approaches !

Rational Speech Acts (RSA) is an information-theoretic, computational framework for making quantitative predictions about pragmatic inferences in context (Frank & Goodman, 2012; Goodman & Frank, 2016). In principle, the pragmatic communicative behavior of reference games is a key candidate for modelling with RSA frameworks, but degrees of flexibility and the issue of scaling to open classes of descriptions and the corresponding need for flexible semantics may prove challenges.

The basic idea of the RSA family of models is to picture two interlocutors recursively reasoning about how the other would produce or interpret utterances, grounding out in a listener (or speaker) who behaves in a pre-specified “literal” way.

Computational frameworks such as RSA provide a way to factor together different trade offs and determine their relative weights in a model (Goodman & Frank, 2016). A softmax is taken over the scores of the options to produce a distribution of interpretations and utterances, with some parameters such as the degree of optimality fit based on data.

This framework is usually run with one or two levels of recursion, where it tends to produce a reasonable fit to human experimental judgments, consistent with work finding that most people reason pragmatically at a low recursion depth (Frank & Degen, 2016).

RSA models have been used to predict some instances of ad hoc pragmatics as well as conventionalized pragmatic implicatures (Goodman & Stuhlmüller, 2013; Bergen?; degen20200406?). Models generally include a utility or informativity term that relates to how well an utterance resolves uncertainty in favor of the target referent. It is common to also include factors such as the prior likelihood of referring to each target (salience prior) and some cost on utterances where longer or more complex utterances are penalized (Goodman & Frank, 2016). Some models also

go beyond informativity, incorporating options to infer the question-under-discussion (Kao, 2014; Qing et al., 2016) or for speakers to balance informativity with politeness (E. J. Yoon et al., 2018).

A full RSA model would incorporate all of these components and infer their weights. However, for tractability, usually only those features that are considered relevant to the domain of interest are included. Because of the flexible framework, it is possible to model many sources and levels of uncertainty, and then integrate out that uncertainty to make predictions, but also update on the sources of uncertainty in response to input.

The Challenge of Semantics

Perhaps the largest challenge to RSA models is the question of how to ground out the models in a “literal” listener or speaker. For the most part, RSA is tested in toy domains where the set of possible utterances are small and it is possible to enumerate a set of meanings (Frank & Goodman, 2012; Goodman & Stuhlmüller, 2013; **bergen?**). For instance, in some domains, a soft or continuous semantics is used to represent that some dimensions of meaning might be more strongly informative than others (**degen20200406?**). This semantics supports the prediction of patterns of “redundant” color adjective use in referring expressions.

Soft semantics can run into conflict with compositionality: either every possible utterance must independently receive a degree of match with every possible object in the prior, or the prior needs to include rules for how to determine the match of a whole utterance on the basis of the match with each component. In a later experiment of (**degen20200406?**), typicality effects made compositional semantics not work, but the utterance space was small enough that each utterance could be treated individually.

In less toy domains, there is not a satisfactory answer: some situations can be handled by empirically measuring likelihoods in an exhaustive ways, but this holistic approach is not compatible with incremental RSA (Cohn-Gordon et al., 2018) or larger sets of utterances that require compositionality to be defined. Some work has used grounded neural language models for alternative generation and likelihood measures, which has scaling potential in domains that have the right grounded datasets

(Cohn-Gordon et al., 2018; Monroe & Potts, 2015; White et al., 2020). In order to extend RSA towards more realistic and open-ended scenarios, an important question to grapple with is what form of meaning (even at a computational level) will appropriately support pragmatic reasoning.

%TODO FOR LATER re-read the neural RSA stuff

RSA approaches to reduction

not entirely sure where this section goes!!

Scaling up RSA to handle reference games requires solving at least two problems. One is specifying a semantics system that can handle the abstract, metaphoric, and parts-based descriptions that are used – this could be seen as the problem of accounting for initial reference.

Secondly, one must explain how to go from one successful utterance to a different (often shorter) utterance. One RSA-style model that attempts to explain why multi-part descriptions are produced initially, but shorter descriptions are produced later is CHAI, a framework to bridge different levels of convention formation (Hawkins, Franke, et al., 2021).

CHAI incorporates parameterized hierarchical uncertainty over lexica that allow for variability in how words will be interpreted that can be integrated over. Listeners’ lexica are not fully known which accounts for background differences that may be shared by a sub-population and differences from communication history that may be individual-specific. The results of inference update these priors, and propagate the new information to both the individual and group lexical representations. CHAI can qualitatively account for various convention-formation phenomena in toy systems (Hawkins, Franke, et al., 2021).

CHAI’s explanation for reduction is consistent with the ambiguity arguments of Piantadosi et al. (2012): initially the context is not sufficiently constraining to resolve the ambiguity of a short utterance, so multiple descriptive pieces are needed to triangulate the meaning, but as conversational history accrues, the context is sufficient to disambiguate the shorter description.

In toy models of interlocuters playing a reference game with soft semantics, initial utterances use multiple properties to collectively increase the degree of certainty in

the target. This successful reference then shapes the prior over word meanings, until the degree of certainty afforded by only one word is sufficient to pick out a referent.

The CHAI framework seems promising in that it qualitatively produces some of the patterns that have been most difficult to explain, namely why it would make sense to change descriptions when nothing in the context has changed. CHAI identifies that something in the context has changed – the conversation history has evolved and changed the word-meaning pairings in the speaker and listener lexica. This solves the issue of presenting a plausible and somewhat testable theory. It leaves open a lot of implementation questions about how to scale it up to be able to interact with (non-simulated, open vocabulary) data, and the model is likely to have a fair number of free parameters so it’s unclear how to test it stringently.

2.4 Phenomena

The communication and conversation literature, especially with regard to referring expressions, has provided useful descriptive characterizations of how people communicate in naturalistic and experimental settings. There is an accumulation of fairly consistent and reliable findings of reduction, convention, and partner-specificity. These describe consistently observable trends, but the necessary and sufficient conditions and the dose-response relationships tend to lack specification. These theories provide a set of potential targets for computational models, although in many cases, more work to clarify the target phenomenon is needed.

It is difficult to directly test these theories as they often fail to make explicit predictions about what would happen under conditions other than those tested or what experiments would adjudicate between theories. Many of these theories present certain results as “special” but our question as we go will be to what extent these phenomena could be explained by the broader coverage theories of efficiency and rational communication.

2.4.1 Mentalizing or non-mentalizing

One historically important distinction that recurs with conversation, and interactions between agents more generally, is whether and how agents track other agents internal

states of knowledge and how this factors into their interaction. While intuitively an important distinction, it's not necessarily a well-formed one. Is this asking whether model-based or model-free approaches capture observed phenomena better? In the present with pretty cool AI systems, there's an issue that appearances and internal architecture don't always align. We can have black-box systems that seem to match rule governed behavior.

2.4.2 Mentalizing and common ground

The “mentalizing” tradition treats humans as representing other humans as agents with internal states that include knowledge and goals. Within this broad school, there is variation in how these representations are implemented, how information gets added or modified, what exactly is tracked, and when representations (versus heuristics) are used.

Within this tradition, many use the term “common ground” to refer to knowledge that two agents share. In some cases, it is used in a pre-theoretic way to mean roughly “things you think another person will understand and won't be surprised if you reference” (Garrison et al., 2022; Leung et al., 2023). For instance, Hanna et al. (2003) defines common ground as the “mutual knowledge, beliefs, and assumptions” held by the interlocutors. This meaning is roughly comparably to “givenness” in other domains (Fay et al., 2010).

However, the problem with the term “common ground” is that some use it in a theoretically very loaded way, originating from the privileged versus mutual versus common knowledge framework (Clark & Wilkes-Gibbs, 1986). Under this usage, “common ground” is defined via infinite recursion in knowing that the other person knows that the first person knows that ...; this is the usage that comes up in formal semantics where many things may be introduced to common ground via accommodation (Horton & Keysar, 1996; Pickering & Garrod, 2004). In practice, humans don't tend to do more than a couple layers of recursion in their pragmatic reasoning (Franke & Degen, 2016). Thus, it is generally not important to distinguish knowledge types at deeper recursion levels than mutual knowledge that both people know to be mutual.

A Bayesian inference approach could complicate this by asking whether sharedness is important or whether this is really a way of saying something about your priors on the other person’s knowledge or lexicon are above some threshold of thinking they probably will understand. The question then becomes how mutual this has to be or whether one can use similar mechanisms to when you believe they know something even if you don’t (they know where the key is, even if you don’t).

How do we determine that something is mutually known with another person? Many approaches have tried to characterize when something is mutually known (Brown-Schmidt, 2012; Clark & Wilkes-Gibbs, 1986; Horton & Keysar, 1996). This has predominately taken a deterministic approach with rules such as “if it’s in shared visual presence, it’s mutually known”. Enumerating all the options doesn’t work because we don’t always have certainty around what is or isn’t mutually known; someone can look at something in shared visual presence, but not know about it because they weren’t attending. Visual presence may be a good heuristic, but good heuristics won’t be perfect, and humans operate under uncertainty. Understanding human communicative behavior also doesn’t require a deterministic answer to when something is mutually known: what another person knows or can be expected to understand is something that computational models will want as an input or intermediary, so that it can be used to evaluate utterance options. However, the knowledge state can clearly be probabilistic and may be inferred from empirical data. Bayesian approaches are great at uncertainty!

%TODO FOR LATER Gabe Doyle stuff

%TODO FOR LATER connect to the lexical side

2.4.3 Interactive alignment theory

The mentalizing approaches can be contrasted with “interactive alignment theory” which attempts to explain how people can successfully collaborate on reference tasks without reasoning about each other’s mental states (Gandolfi et al., 2022; Pickering & Garrod, 2004). The motivation for non-mentalizing accounts is the apparent difficulty of mentalizing, coming from accounts such as naive egocentrism, naive realism, and initial egocentrism [Veronica promises citations for this, but later]. %TODO FOR LATER These intellectual traditions claim that mentalizing is a mentally taxing

add-on, that is computationally expensive and not automatic. Under this framework, (Gandolfi et al., 2022; Pickering & Garrod, 2004) try to account for observed alignment between interlocutors by extending ideas such as lexical alignment [Veronica again promises citations] TODO FOR LATER Pickering & Garrod (2004) claims that the alignment occurs via “priming” and is “resource-free and automatic”, without providing a further explanation of what this means or how this works in terms of memory and processing. Given that humans reason socially about each other readily and from a young age (Rakoczy, 2022), it’s not clear how well the motivation for a non-mentalizing approach holds up. An expansive view of mentalizing could also incorporate more heuristic patterns that don’t involve “explicit” inference and some of what looks mental could also be repetition of ease of use underneath (cite macdonald). However, as expressed, the alignment theory doesn’t provide a algorithmic or mechanistic level theory.

One additional complication is convention formation requires not just alignment but specific patterns of alignment where the utterances become shorter rather than just mimicked. To my knowledge, interactive alignment does not have an explanation for this, in part perhaps because their studies tend to be of a different sort. The “common ground” tradition generally using asymmetric director/matcher designs (dating back to at least \citet{krauss1966}) and the “interactive alignment” traditions using symmetric designs such as the ‘maze’ task. Thus it is possible the two approaches are build around trying to explain differing sets of experimental results.

2.4.4 Partner specificity

One key phenomenon from iterated reference games to unfamiliar images is that switching matchers or adding a new matcher changes the describers behavior, as they shift to longer descriptions. This change in behavior is described as “partner-specificity”, with the idea being that the conventional names developed with one partner as specific to that partnership (Brennan & Clark, 1996; Hawkins, Liu, et al., 2021; metzing2003a?). The idea of partner-specificity is also referenced with regard to how different pairs diverge to different names for the targets (hawkins2020b?). Partner specificity is part of the mentalizing tradition and assumes that partners (and their background and knowledge states) are being represented in the minds

of speakers. The form of the representation is not explicitly stated, so these models could be compatible with heuristic or distributed representations, but include explicit thinking about the audience and are incompatible with the non-mentalizing “priming” account.

Empirical evidence from experiments where one director talks with multiple partners suggests that people do “partial pooling” over their partners (Hawkins, Franke, et al., 2021; S. O. Yoon & Brown-Schmidt, 2014). That is, a speaker A will show some variation in their expressions when talking to partner B versus partner C, but there will be some generalization between partners as well, so that A talking with B is more like A talking with C than D talking to E. When coupled with a tendency for descriptions to shorten within a pair, this leads to a jagged pattern of reference length: when switching to a new partner, speakers use longer utterances, but not as long as their initial utterance with their first partner (S. O. Yoon & Brown-Schmidt, 2019b).

This pattern could be efficient if the context from the initial utterances is requisite to understand the later parts (we will empirically address this question in Chapter 4). The jagged pattern in particular is suggestive of a mix of speaker production effort and shaping the matcher’s lexicon.

Partial pooling and partner specificity can be explained at least theoretically at a high level by RSA family theories including CHAI and lexical uncertainty, although computational and semantic tractability problems prevent testing this with natural language as of yet.

2.4.5 Audience design

A related term is “audience design”, the idea that speakers seem to be sensitive to the knowledge state of their listener and say things that are easy for the listeners to comprehend. Confusingly, “audience design” sometimes implies intention on the part of the speaker (Horton & Gerrig, 2002, 2005), and sometimes is used when utterances are constructed based on what’s easy for the speaker, and listener ease is a side effect (Horton & Keysar, 1996; MacDonald, 2013; Rogers et al., 2013). For instance, audience design could both occur in times when the speaker gives an elaborated description to a naive listener (inferred to be intentional if contrastive with their

description to a non-naive listener), but speakers may also tend to start descriptions with given material which is both more accessible to the speaker and convenient for the listener. Intention versus side-effect are difficult to distinguish between because speakers and listeners often share recent context, find the same things salient, and linguistically what is easier to produce is often easier to process. Thus, disentangling speaker and listener ease may require careful experimental designs where ease of production and ease of comprehension are separated (Ferreira, 2004).

2.4.6 Effort sharing

Questions around audience design are related to larger issues of how interlocutors split the communicative burden with one another. Depending on the task and the communication modality, there may be many options for how to balance the communicative load (Clark & Wilkes-Gibbs, 1986; Fay et al., 2010; Fox Tree & Clark, 2013). For instance, a listener could describe what options they see or otherwise prompt the speaker. We might expect the load splitting to vary based on the capacities of the interlocutors (ex. a speaker might craft their utterances more when talking to a child versus an adult) and the capacities of the channels (ex. speakers may use different approaches if listeners can interrupt).

Multi-way conversations complicate the verbal theories of audience design and partner specificity by introducing a larger audience of more partners. Two main questions are whether “aim low” or “aim high” in balancing the needs of the listeners and whether speakers track individual listeners or an aggregate (S. O. Yoon & Brown-Schmidt, 2014). Empirical results indicate that speakers are sensitive to the knowledge states of listeners in a gradient way (S. O. Yoon & Brown-Schmidt, 2014, 2018, 2019a). At least in small groups, speakers can track the correspondence between individual listener identity to histories and knowledge states, and can incorporate contextual factors that modulate task difficulty into their considerations (S. O. Yoon & Brown-Schmidt, 2019b). Speakers also take strategies in group contexts that don’t occur as often in dyadic contexts, such as referring to a target with both the name that one person will understand and a elaborated description that will help another person get on the same page (S. O. Yoon & Brown-Schmidt, 2018). The ability to track partner’s knowledge states presumably would degrade as groups got bigger,

but this paradigm has not been used for groups large enough for this to happen.

Heirarchical Bayesian models or noisy models could account for gradients if tracking mutliple people leads to fuzziness, it could be possible to place inference limitations to get a noisy version of the full system rather than require either full inference or heuristics. (lol maybe cite Ben’s cogsci)

2.4.7 Convention formation

Over repetitions with the same partner, dyads in repeated reference games tend to form shared “conventions” (also called “conversational pacts”) about how to refer to the initially ambiguous targets. These conventions tend to be partner- and context-specific: changes in the speaker, audience members, or changes in the context can all license the use of a new description (Ibarra & Tanenhaus, 2016; S. O. Yoon & Brown-Schmidt, 2014; **metzing2003a?**).

What exactly does convention formation refer to? There is ambiguity about what level of specificity convention formation and conceptual pacts refer to. It could be on the lexical level, such as calling a figure “ballerina”. It could be conceptualizing the figure as a ballet dancer with a tutu (manifesting in descriptions with semantic association, but not lexical overlap, such as “ballerina” and “dancing in a tutu”). It could also be a general paradigm for how to describe figures, such as in terms as humans in different postures. \citet{horton2002a} distinguish between “lexical entrainment” when the same words are reused, and “conceptual similarity” when there is broader similarity that does not repeat the same words. These levels often co-occur, but in order to have a computational model of the phenomenon, we need to be clear about which is meant in order to operationalize it.

A heirarchical Bayesian model could potentially be used not just for partially generalizing over interlocuters but also for partial generalization over targets. At the very least, an interlocuters acceptance of “ballerina” for a figure implies that their lexicon might also accept other semantically related words (“dancer”). Should we think of lexical uncertainty as a warp over not just individual words by semantic spaces?

2.4.8 Reduction

The semantic meaning of a description is not a priori related to its length, but these two features empirically tend to correlate in iterated reference games (**hawkins2020b?**). Thus “reduction” or the shortening of utterances is sometimes used as a shorthand and measurement proxy for the semantic changes (Clark & Wilkes-Gibbs, 1986; Hawkins, Franke, et al., 2021). Convention and reduction are sometimes conflated with partner-specificity, as these phenomena often co-occur with different pairs forming different conventions and changes in group composition leading to (temporarily) longer descriptions (Clark & Wilkes-Gibbs, 1986; Wilkes-Gibbs & Clark, 1992).

It remains an empirical question whether the shortening of utterances, convention formation, and partner-specificity of descriptions are inseparable or merely occur together in the experimental paradigms considered in the literature.

TODO there’s an efficiency angle here?

Testing these patterns would require studying the phenomena in more varied experimental situations. One question is how convention formation might change with group size, as some of the pieces, like partner-specificity seem like they will have to fall apart if groups grow to dozens of people. One angle on larger groups is network structures, where (non-linguistic) convention formation has been studied on networks of up to 50 people (Guilbeault et al., 2021). In this communication game with targets varying over continuous space, large groups tend to end up with fairly consistent category boundaries across independent networks, while small networks can support more idiosyncratic category boundaries. This work is suggestive that there may be group-size and group/network-structure dependencies on what sort of conventions arise. There has been some work on group and network designs in traditional iterated reference games, but only on relatively small groups.

Typically, in convention formation, holistic or analogic descriptions are the ones that stick (Clark & Wilkes-Gibbs, 1986), but this isn’t absolute. Groups can successfully coordinate on reference using many different types of names, including ones that pick up on low-level or meta-level features. The range of successful options makes explaining convention formation harder, and suggests suggests a strong path dependency for how reduced utterances evolve, potentially influenced by factors such as relationships and humor value.

2.4.9 Developmental work

Redo this because we know more now from the paper!

Many of the phenomena discussed above are likely to depend on varied communicative and linguistic skills. One question therefore is at what age these phenomena emerge, as that provides insight into when the underlying skills must exist by. There has been relatively limited work on children, and much of it is hampered by small sample sizes and limited numbers of experiments. Thus, unlike the robustness of the key phenomena in the adult literature, there is uncertainty about the generalizability of the findings with children.

Children show rudiments of the key phenomena from an early age, although they remain unadultlike for a while. Children ages 3 to 5 show sensitivity to referential pacts, in protesting the breaking of a pact, although they sometimes protest even when a new speaker uses a new term (Matthews et al., 2010), in contrast to adults who allow new speakers to form new pacts (**metzing2003a?**). In a test of both context and partner-specificity (modeled on Brennan & Clark (1996)), 6 year olds showed the adult-like pattern of maintaining the same terms with a partner even when a context change rendered them overinformative, while 4 year olds struggled more on the task and did not show partner-specificity (Köymen et al., 2014). In addition to partner-specificity, adults are also sensitive to the context and whether distractor items are close or far. Evidence for children is mixed, with Abbot-Smith et al. (2016) finding some signs of sensitivity to context in 2 and a half year olds, but in a different paradigm 4-8 year old children did not appropriately modulate labels to familiar objects based on context (Leung et al., 2023).

In repeated reference games to novel objects, there are claims of young children's inability to form referential pacts, followed by a gradual increase in competence throughout childhood and adolescence, although these studies have methodological concerns that could mask earlier competence (Glucksberg et al., 1966; Glucksberg & Krauss, 1967). More recently, 8-10 year old children in Branigan et al. (2016) seem to demonstrate an ability to describe abstract shapes and some level of convention formation (measured via reduction), although children show mixed results in terms of sensitivity to an existing versus naive participant (compare adult performance in Wilkes-Gibbs & Clark, 1992).

Overall, the available literature indicates a gradual path towards adult-like behavior, but the limited number of studies means all these claims should be interpreted with caution. Understanding the developmental trajectory of pragmatic and communicative skills requires more rigorous and systematic data on children.

How does development factor into the models? How much do computational models make claims about kids? From an RSA type standpoint, we could cache out various pieces that could be changing across development – children’s lexicon and access to word options could vary, the number of draws they get and inference cost could vary. One not yet well developed portion is how to handle multiple utterances within the same referent in RSA – how does a matcher decide when to comment or ask a question versus make a selection (and if so what to say?). It’s certainly possible to tweak parameters in ways that could mimic what is known of children. Also could up the temperature, but we don’t really know.

2.5 Processing

2.6 Naive matcher stuff

2.7 Summary

The communication and reference game literature provides descriptive theories that identify some phenomena of interest and raises questions around whether these phenomena occur intentionally or emerge as a by-product of other processes. Within the narrow range of experimental paradigms, clear patterns around reduction, convention, and partner-specificity are robustly observed, and could form a list of results that models could hope to explain and predict. Our current understanding of these phenomena is limited. We do not have ways to predict them quantitatively, nor do we have a grasp on the necessary or sufficient experimental conditions under which they occur (to what extent). In many cases, we do not even have consensus on how exactly the characterize the phenomena. In order to build strong theories, we first need clarity around what exactly the theories should be accounting for. A core theoretical question is whether the observed patterns of reduction, convention, and partner-specificity

require “special” mechanisms, or whether these results can sufficiently be explained by broader coverage theories of efficiency and rational communication.

2.8 Where does this leave us?

2.8.1 Efficiency

Sometimes, judging whether something is efficient or not may be clear cut, if all reasonable sets of assumptions return the same result. It is perhaps easier to judge something to be inefficient if there is a shorter alternative that is both easier to produce and easier to comprehend. In the general case, however, efficiency is very hard to cache out in specific predictions because of the many time scales the pressures operate on. What’s efficient for an utterance in isolation may not be efficient when considered over an entire life of language use. Thus, the efficiency framework is not directly testable, but it’s goodness as a theory instead relies on the parsimony of the linking theories that are required to meld it to the data.

The formation of conversational pacts is often characterized as efficient, but this claim has not been cashed out in formal models like those described above (Clark & Wilkes-Gibbs, 1986; **hawkins2020b?**). “Efficiency” is an informal description of these pacts that captures that shorter expressions are *more* efficient, but a given shorted expression could still be mis-calibrated to a particular situation. It could be too short, leading to misunderstanding, or still longer than it needs to be for effective communication. Mis-calibration of this type could contradict claims about partner specificity of reduction, but would not have to. Communicators could intend to be optimally calibrated but run into production difficulties, masking their competence. Either way, current claims of efficiency in conversation do not connect directly with the formal claims of optimal efficiency in the “language design” literature, and it is unknown whether speakers and listeners are calibrated or not.

Key questions this leaves for us: where do we think the bottle necks are in real time? inference versus saying more? what about goals other than communication/info delivery? how do we factor efficiency in there? (even if we semi control the task, it’s still people interacting and there’s a lot about communication norms) but we can test at least narrow pieces

2.8.2 Bayesian

Two potential problems are scaling up to an open vocabulary and free parameters that will lead to high flexibility and a lack of risky prediction (Meehl, 1990). The open vocabulary problem is a hard one; rather than wait for a fully realized semantic system, I think one criteria for judging the adequacy of semantic systems is whether they can serve as a linking hypothesis to allow RSA models to predict the patterns of pragmatic language use that are observed experimentally. CHAI, while not a full explanation for the full patterns of real-world data, is a big step forward in at least providing a framework that actually explains why reduction would be optimal. It remains to be seen whether RSA-style models can predict the slope of reduction and the content of words that stay versus drop, but I believe the attempt would push forward our understanding of pragmatics and communication. Even without a cached out model, this approaches can inform how we think about experiments and results.

Chapter 3

Old

3.1 Psycholinguistic considerations

The utterances in reference games that optimization-oriented theories seek to explain are the product of lower-level, incremental processes. The algorithmic level of linguistic communication is constrained by its instantiation in the mind. To the extent we can infer these constraints from fine-grained behavior or transfer these constraints from other areas of psycho- and neuro-linguistics, these constraints may provide bounds for the computational models or provide testable predictions at these other levels of analysis.

%TODO FOR LATER mike is dubious about this for so-called “computational level” models or something

%TODO FOR LATER maybe cite the new shain paper as an example of how different levels influence each other

3.1.1 Top-down or bottom-up

One large question in language processing and production broadly is what the relative balance of bottom-up and top-down influences are (Gwilliams et al., 2022; Horton & Gerrig, 2005; Horton & Keysar, 1996; Tanenhaus et al., 1995).

In the psycholinguistics of reference games, a hotly debated issue is whether the early moments of production and processing are “ego-centric” or can be influenced

by non-linguistic information, such as the perspective of the interlocuter. On the production side, Horton & Keysar (1996) attempted to test this theory by comparing utterances produced with and without time pressure. They interpret the apparently ego-centric utterances in the speeded condition as evidence that initial utterance planning is ego-centric, but that monitoring and fixing of the utterance may take into account the listener's perspective, and may occur prior to utterance initiation.

On the comprehension side, Keysar et al. (2000) argued for an initially egocentric perspective on the basis that people often initially look at objects that are good matches to a description even if the object is not mutually visible to the speaker. This interpretation rests on a couple dubious linking hypotheses: that if people consider an interlocuter's perspective, their prior should be that the interlocuter only refers to mutually known things; and separately, that looking at an object is a sign that it is considered a potential referent (eye-tracking data is widely interpreted this way, but there is evidence that this proxy is only approximate, **degena?**).

The counterpoint to initial ego-centrism presented by (Hanna et al., 2003) is a constraint-based theory where many factors can play into comprehension, including working memory limitations. Many factors may influence language production and comprehension, to varying extents, and on differing time courses. Determining the relative timings and weights is an important endeavor that will require careful experiments over a systematically varied swatch of experimental space.

As these experiments show, it's empirically difficult to do so when the measurements are far from the constructs and there are many nuisance variables to abstract over. An experiment compares two conditions with different objects visible to the speaker but not the listener (or vice versa), and can claim that for this set up and these objects, people look at the privileged object some amount. Is looking at the object predicted only by ego-centrism theory or can it also be predicted by some other pattern such as the constraint based approach? When a different experiment with different stimuli has a different result, is this because of nuisance differences in the stimuli? Or because of critical differences in how well each stimuli matched the verbal description? Even if we had well-characterized descriptive work about when this early-reaction occurs in terms of stimuli and conditions, there's still an interpretive question of what the early looks mean.

Some of the experiments around egocentrism try adding time pressure to get at earlier stages of production, but we don't actually know how time pressure interacts with the production system – does time pressure cause people to short-cut off the end, or to satisfice more at all the stages?

Perhaps more fully fleshed out versions of initial-ego-centrism theory and its alternatives and their requisite linking hypotheses could define a set of experiments where they each make clear and differing predictions. For now, with the existing theories and experiments, interpretations lie on a bunch of promises about assumed (untested) generality and linking assumptions.

3.1.2 Production constraints

In addition to possible information-integration timing limitations on optimality, the retrieval and generation of utterances may be another source of deviation from optimal models. Utterance planning is difficult, and production biases such as easy first, plan reuse, and reduce interference may produce deviations from information-theoretic predictions (MacDonald, 2013).

Another limitation is the search problem of production. RSA and other computational theories assume the existence of alternatives and then provide ways of choosing from among the options. Especially with low-codability targets, the initial mental generation of any potential referring expression may be a bottleneck. There are empirical challenges with determining what is difficult for speakers to produce, although analyses of disfluencies is one approach (S. O. Yoon & Brown-Schmidt, 2014). Getting traction on initial utterance planning could usefully inform the generation of alternatives in information-theoretic models, but this is likely very challenging.

3.1.3 The need for different kinds of data

From a production side, we might want to know what constraints are how influential on speakers' utterances (for instance, wanting to know if they are initially egocentric). If all we have is the transcript of what is eventually produced, there's not much we can do. We'd have to make a lot of assumptions about how the time course of production maps to a final utterance. Maybe we can say something based on what

types of phrases are produced at the start of an utterance versus the end. But this still assumes that the time scale of articulation reflects the timescale of planning. Which is often true, but assumes that speakers don't pause to fully plan their utterance before beginning to emit the utterance.

Timed data could address this by revealing where there are (filled or unfilled) pauses, and how long speakers take before initiating the utterance. This is still an indirect measure of what is happening in the mind, but it gives richer data for theories to fit to and is thus a stricter test.

We still can't know even from this data whether there's a very early initial stage that works somehow but is overcome before any utterance exits the mouth. Here, we would want other measures, such as possibly eye-tracking or neural data to look for evidence of what is being thought about prior to utterance initiation. As a caveat, interpreting these signals will still rest on a number of assumptions.

For the comprehension side, accuracy data provides the least constraint and insight into the process, reaction time provides a little more, and eye-tracking or other continuous measures may provide more, although they still require linking assumptions.

3.1.4 Takeaways

Eventually models and evidence at all levels need to converge for a satisfying network of theories. While many of the phenomena of interest are described at a fairly high level in terms of observed utterances, the mechanisms by which these utterances and interpretations are produced go through the language processing and production system.

In some cases, the questions of interest are about the fine-grained time course of reference games: how utterances are generated and interpreted and what information is integrated when. In other cases, the theories rely on implicit assumptions, such as ideas that the language system does something that can be approximated as generating and evaluating alternatives. These assumptions and questions are about the the general processes of language cognition which this instance of language use shares with other instances of language use.

Even if they are not the target, psycholinguistic considerations need to be part

of the parsimony of theories. High-level theories write promissory notes that at least algorithmic approximations will be found for linguistic or memory processes – we should seek theories that make promises that are most compatible with psycholinguistic findings and theories.

3.2 Ways forward

%What would we want from formal theories and computational models?

%Any theory also needs to be compatible with what we know at different levels of analysis and related phenomena. A theory of linguistic iterated reference to novel objects need not cover how named objects are referred to, or how conventions form in drawing, or how language evolves, or how language is processed in the brain. But it is important the theory be compatible with all of these; it should not make untrue assumptions about these other areas. Ideally, the boundaries of a theory would be continuous with adjacent results and theories.

%Related, there are questions of how to scope a theory of iterated reference games; for instance, it may make sense to have a model that only covers dyadic interactions or a certain modality of interaction or a certain type of stimuli. This limited scope is fine if it is clear and the assumptions of the models don't lead to wild (false) predictions.

A key part of formal theories and computational models is they need to make clear

Before we can do that, we need is a better definition of what is to be predicted.

The gap between data and theory can be narrowed from both the data and theory sides, so I lay out a few potential avenues for progress.

3.2.1 Data side

One necessity is a better empirical and descriptive understanding of the phenomena

We don't have to finely map the entire landscape, but some sense of the geography

Another empirical avenue is to fill in descriptive details on the process of reduction.

Another third approach is to push down a level of analysis and do empirical work looking

3.2.2 Theory side

In the introduction, I listed three scales on which one might try to make theories%TODO : the scale of an entire conversation, the scale of one instance of referring, and the s I think the most viable of these is the second, where we try to get predictive or causal

This scale is the closest to having formal models available, in the form of RSA-style mo

%Basically, given a history to a certain point, how to predict what interpretations or u Two big questions here are how to handle the open-ended semantics and how to construct t

%TODO FOR LATER ugh this is getting away from RSA though

%The big questions here are how to handle open-ended semantics and how to construct the

Another direction where RSA models don't have the capacity to match the type of data fro

The other main theoretical approach is the efficiency angle. Unfortunately, efficiency m

Current statistical models used to assess reduction (or disfluency rate, etc) have t

3.3 What would it take to have a (satisfying) computational model

- why no computational model Veronica? -> it's pretty rich and complicated and parts of this make testing a model at scale hard (and creating a testable generative-y model hard)

- am interested in thinking about that, but I think we can start by testing

- some bounds on the (verbal) theories to get a lay of the land
- also sets up what ways a model needs to be expandable ()

3.4 Outline of the dissertation

In this dissertation, we expand the empirical foundation of iterated reference games in three directions, with the aim of addressing some theoretical assumptions and questions and providing some data for computational modeling in the future. We focus on the microcosm of iterated reference games, apply a wider angled lens. I hope that this work better characterizing the phenomena of interest across contexts and levels of analysis provides some ground-work for future theory development.

In Chapter 2, we address the generalization patterns of the key patterns of results in iterated reference games, by examining the trajectories of convention formation in games of varying small group sizes and varying communication channel capacities. This expands the empirical foundation to mimic more of the different circumstances in which communication occurs in the real world, and allows for a test of how entangled the different reduction phenomena really are.

In Chapter 3, we turn to the developmental timeline of the communication and coordination skills embedded in iterated reference tasks. Here we aim to re-visit historical claims that children are too egocentric to produce successful ad-hoc referential expressions, which is fairly incompatible with more recent work showing nascent pragmatic language use and communication skills in preschoolers. This different context again provides a test of how entangled different observed reduction phenomena really are and provides insights into how language use fits into broader cognitive and linguistic development. Finally, in Chapter 4, we measure the level of opacity in the referential expressions produced in iterated reference games (using the corpus of reference games from Chapter 2). Taking inspiration from lexical uncertainty models, we use both naive human matchers and computational proxies to determine how far from “baseline semantics” referential expressions are at different stages of convention formation. These patterns are a test of at least the strongest versions of efficiency and audience design claims and give us a better understanding of what sort of partner-specificity is at play.

? Are we including processing results??

In the last chapter, I conclude with a discussion of where this leaves us and what directions seem most promising from here.

3.5 For conclusion

readdress Current statistical models used to assess reduction (or disfluency rate, etc) have the issue that they are fit to a specific dataset and can't be compared across datasets and situations, so it's difficult to assess their quantitative generality (Yarkoni & Westfall, 2017).

3.6 Misc

Our understand and predictive power around these phenomena is limited. We do not know how they respond to “twiddling the experimental knobs”, nor are there theories that make risky, testable predictions about necessary and sufficient conditions or the functional form. We can describe the trends of what happens in the range of conditions that have been experimented on, but theories rarely make predictions of what would happen in other conditions or what experiments would adjudicate between theories.

%However, our understanding of these phenomena is limited. We do not have ways to predict them quantitatively, nor do we have a grasp on the necessary or sufficient experimental conditions under which they occur (to what extent). In many cases, we do not even have consensus on how exactly the characterize the phenomena. In order to build strong theories, we first need clarity around what exactly the theories should be accounting for.

%A core theoretical question is whether the observed patterns of reduction, convention, and partner-specificity require “special” mechanisms, or whether these results can sufficiently be explained by broader coverage theories of

At whatever level, we would like to eventually have theories that are precise in wha

I will cover approaches from four perspectives. First, the communication and reference g

One can define the question of interest, and thus the scope of the modeling and prediction problem, at a few different levels:

1. A clear trend observed in the iterated reference games is that utterance length
2. Instead of predicting the entire curve of a conversation, we could instead ask:
3. Finally, one may wish to model the actual production and comprehension processes

%TODO FOR LATER Mike would like a diagram or table or **something**

Currently, we have formal models at none of these levels. Personally, I find the 2

%
 % Hawkins et al. (2020) How do you break symmetries in initial descriptions: there's prior variability across speaker preferences (they may each have a preferred label, but be unsure if others will accept it) and/or speakers may also not have labels and need to do some sampling. This doesn't account for production time course factors. %
 % Hawkins, Franke, et al. (2021) says there are 3 core cognitive abilities: the ability to represent that there is variability in other's lexicons; to coordinate via online learning; and to generalize across interactions ("partial pooling" model where updates both to partner and population) % %Efficiency also predicts that a changing conversational history will change the context and thus different descriptions may be efficient. This could operate both by increasing beliefs that a certain utterance will be understood (and this is contextually low ambiguity) or more generally by shaping the syntactic expectations, perhaps making it easier to produce and comprehend odder descriptions. % %Hawkins, Franke, et al. (2021) separating the inference problem about what the other person's lexicon is (which is how they will interpret things in the moment, b/c you may have temporarily changed their lexicon) with decisions about what to say given that % %Hawkins, Franke, et al. (2021) points out that our models for communication and modeling the world and others need to be able to account for different people having different knowledge (including some tied to community

membership or social role) and that vocabularies need to accomodate change over time and new things to refer to as the world changes.

%TODO FOR LATER FIX REFERENCES!!!

Chapter 4

Interaction structure

Placeholder

4.1 Introduction

4.2 Results

4.2.1 Overview of experiments

4.2.2 Smaller and higher-coherence groups are more accurate

4.2.3 Smaller and higher-coherence groups are more efficient

4.2.4 Larger groups make greater use of matcher contributions

4.2.5 Descriptions converge faster in groups with thicker channels

4.2.6 Games with thicker channels diverge from one another more quickly

4.3 Discussion

4.4 Materials and Methods

4.4.1 Participants

4.4.2 Materials

4.4.3 Procedure

4.4.4 Data pre-processing and exclusions

4.4.5 Modelling strategy

Chapter 5

Developmental origins

Placeholder

5.1 Introduction

5.2 Experiment 1

5.2.1 Methods

Participants

Materials

Procedure

Data processing

5.2.2 Results

Accuracy and speed

Description length

Convergence

5.2.3 Discussion

5.3 Experiment 2

5.3.1 Methods

Participants

Materials

Procedure

Data processing

5.3.2 Results

Accuracy and speed

Description length

Convergence

5.4 Joint analysis

5.5 General discussion

Chapter 6

Opacity and processing

Placeholder

6.1 Introduction

6.2 Task setup

6.2.1 Materials

6.2.2 Experimental procedure

6.2.3 Computational models

6.3 Experiment 1

6.3.1 Methods

6.3.2 Results and discussion

6.4 Experiment 2

6.4.1 Methods

Experiment 2a

Experiment 2b

6.4.2 Results

Experiment 2a

Experiment 2b

Additional predictors

6.4.3 Model results

6.4.4 Discussion

6.5 Experiment 3

6.5.1 Methods

6.5.2 Results and discussion

6.6 General Discussion

Conclusion

Wow at some point there will be text that goes here. Isn't that exciting!

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Placeholder

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